



MINING 2.0: ADANI GROUP'S BLUEPRINT FOR SUSTAINABLE, RESPONSIBLE RESOURCE SUPPLY

The fourth-largest economy in the world's mining sector is being transformed—once traditional and contentious, it's now tech-driven and sustainable. Adani Natural Resources is leading the paradigm shift towards a modern, responsible mining future.

The new golden age of Indian mining is upon us. Data shows that from 2010 to 2024, India's annual production value surged from ₹2 lakh crore to ₹4.5 lakh crore, helped by policy reforms and digitised auction systems. Now, the mining sector contributes around 2 per cent to GDP (gross domestic products), employs over 50 lakh people directly, and supports more than 80 lakh livelihoods in ancillary industries.

But despite this outlook, major challenges persist such as fragmented ownership, ageing workflows, and ecological pressures. The demand for innovation has never been more pressing.

Adani Natural Resources, which was formed in 2008 and has its presence in five states — Madhya Pradesh, Maharashtra, Jharkhand, Odisha and Chhattisgarh — is powering transformational changes.

PEKB Becomes Chhattisgarh's First Solar-Powered Mine

In a bold departure from conventional mining, a 30-acre solar array now shines atop reclaimed land at the Parsa East and Kente Basen (PEKB) coal mine in Surguja, Chhattisgarh. Commissioned in June 2025, this 9 MW (megawatt) solar plant, built by Mundra Solar PV Ltd. (a part of Adani Green Energy Ltd.), marks a landmark shift in how India powers its mines.

This is Chhattisgarh's first mine to run on solar energy—and among the select few in India. At a time when mining is still largely fossil-fuel dependent, PEKB block sets a national precedent for energy efficiency

and environmental stewardship.

Over its lifetime, the solar plant is expected to offset nearly 4 lakh tonnes of carbon di-oxide—equivalent to planting 2.5 million trees. Annually, it will prevent over 17,000 tonnes of carbon emissions. At the inauguration, Ambikapur legislator, Rajesh Agrawal, stated: "This solar plant proves that even coal mines can lead with sustainability."

By using mined-out land for energy generation and feeding clean power back into operations, PEKB is demonstrating that coal and clean energy can coexist responsibly when backed by innovation and intent.

Revolutionising Operations with Dozer Push Technology

In another industry first, the Council of Scientific and Industrial Research (CSIR) and the Central Institute of Mining and Fuel Research (CIMFR), in collaboration with Adani Enterprises Limited (AEL), has successfully trialed the Dozer Push Mining Method at the PEKB coal mine. Conducted in Chhattisgarh, this method introduces automated, unmanned mining technology to enhance safety, reduce environmental impact, and increase operational efficiency.

The trial included 108 holes drilled using automated rigs, followed by cast blasting with 60 tonnes of emulsion explosives. The blasted material will be pushed using large, automated dozers—significantly streamlining the mining process.

This innovation stems from a CSIR-CIMFR-led scientific study tailored to Indian conditions over two years,



"At Adani Natural Resources, we are redefining mining with cutting-edge technologies and sustainable practices.

Our commitment to innovation and environmental stewardship is transforming the industry, making it safer, more efficient, and eco-friendly. By integrating hydrogen mobility, automation, and renewables, we are setting new standards in 'responsible mining' to create model mines. The changing face of mining is not only revolutionising operations but also making the sector attractive for young talent, eager to be part of this exciting transformation."

DR. VINAY PRAKASH,

Director of Adani Enterprises Limited and the
CEO of Adani Natural Resources